

**Multiple Choice:**

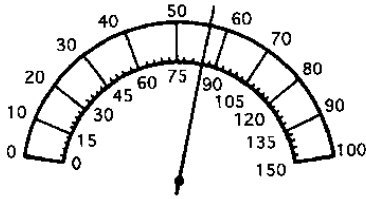
1. As the current range of an analog meter is increased, the overall meter resistance, R_M _____.
A. decreases
B. increases
C. stays the same.
D. None of the above.
2. When set to any of the voltage ranges, a typical DMM has an input resistance of _____.
A. about $0\ \Omega$
B. $10\ M\Omega$
C. $20\ k\Omega$
D. $1\ k\Omega$
3. Defined as the controlling force acts in opposition to the deflecting force and ensures that the deflection shown on the meter is always the same for a given measured quantity.
A. Operating device
B. Controlling device
C. Damping device
D. Linear device
4. What is the method of damping for a moving-coil instrument?
A. Linear
B. Air
C. Eddy current
D. Hairspring
5. The change in the direction of a compass needle, when a current-carrying wire is brought near, is:
A. Electromagnetic deflection.
B. Electrostatic force.
C. Magnetic force.
D. Electroscopic force.
6. The _____ is basically a graph-displaying device.
A. oscilloscope
B. universal meter
C. sweep generator
D. probe
7. For a moving-coil meter movement, I_M is
A. the amount of current needed in the moving-coil to produce full scale deflection of the meter's pointer.
B. the value of current flowing in the moving-coil for any amount for pointer deflection.
C. the amount of current required in the moving-coil to produce half-scale deflection of the meter's pointer.
D. None of the above.
8. Which of the following statements is false?
A. The Schering bridge is normally used on measuring unknown capacitances
B. A.C. electronic measuring instruments can handle a much wider range of frequency than the moving coil instrument
C. A complex waveform is one which is sinusoidal
D. A square wave normally contains the fundamental and even harmonics
9. A $20,000\ \Omega/V$ VOM indicated a reading of 250 volts on its 300 volts scale when used to measure an unknown DC voltage. If a $500\ k\Omega$ resistor was connected in series with the VOM when the test was made, determine the value of the unknown voltage.
A. 271 V
B. 250 V
C. 350 V
D. 550 V
10. When using an ohmmeter to measure resistance in a circuit.
A. the power in the circuit being tested must be off.
B. the power in the circuit being tested must be on.
C. the power in the circuit being tested may be on or off
D. the power in the circuit being tested should be turned on the after the leads are connected.
11. The resistance of a moving-coil instrument is $10\ \Omega$ and gives full-scale deflection at 10 mA. Calculate the resistance of the shunt required to convert the instrument to give full-scale deflection when the circuit current is 5 A.
A. $0.02004\ \Omega$
B. $1\ \Omega$
C. $0.20004\ \Omega$
D. $2.04\ \Omega$
12. The milli-ammeter is converted into a voltmeter by connecting a high value resistance in series with it. This is called _____.
A. series
B. multiplier
C. shunt
D. bypass
13. As the voltage range of an analog VOM is increased, the total voltmeter resistance, R_v ,
A. decreases.
B. increases.
C. stays the same.
D. None of the above
14. An ammeter shunt is useful because:
A. It increases meter sensitivity
B. It makes a meter more physically rugged
C. It allows for measurement of a wide range of currents.
D. It prevents overheating of the meter.
15. To use an analogue oscilloscope, three basic settings to accommodate an incoming signal need to be adjusted. To adjust the attenuation or amplification of the signal:
A. use the time/division control to set the amount of time per division represented horizontally across the screen.
B. use the volts/division control to adjust the amplitude of the signal before it is applied to the vertical deflection plates.
C. use the trigger level to stabilize a repeating signal, as well as triggering on a single event.
D. none of the choices
16. Suppose a certain current in a galvanometer causes the needle to deflect 20 degrees, and then this



- current is doubled. The needle deflection will ____.
- A. decrease
B. stay the same
C. increase
D. reverse direction
17. Which meter has a higher resistance, a DMM with 10M Ω of resistance on all dc voltage ranges or an analog VOM with a 50 k Ω /V rating set to the 250-V range?
- A. The DMM.
B. The analog VDM.
C. They both have the same resistance.
D. It cannot be determined
18. An ammeter with a full-scale deflection of 1 mA has a calibration error of $\pm 3\%$ of the full-scale current. If the ammeter reads 0.3 mA, what is the possible range in the true current through the ammeter?
- A. 0.27 mA – 0.33 mA
B. 0.3 mA – 0.33 mA
C. 0.23 mA – 0.37 mA
D. 0.56 mA – 0.73 mA
19. What type of moving iron instrument has a current that flows in the solenoid, a pivoted soft-iron disc is attracted towards the solenoid and the movement causes a pointer to move across a scale?
- A. Attraction type
B. Repulsion type
C. Pivot type
D. Moveable type
20. Calculate the series resistance needed to convert a D' Arsonval meter with a resistance of 1.5 k Ω and full-scale of 150 μ A into a voltmeter with 15 V range.
- A. 1 k Ω
B. 86.5 k Ω
C. 25.75 k Ω
D. 98.5 k Ω
21. For an analog VOM with a mirror along the printed scale.
- A. the pointer deflection will be magnified by the mirror when measuring small values of voltage, current, and resistance.
B. the meter should always be read by looking at the meter from the side.
C. the meter is read when the pointer and its mirror reflection appear as one.
D. Both A and B.
22. Which of the following would refer to a moving coil rectifier instrument?
- A. An uneven scale, measuring dc.
B. An even scale, measuring a.c.
C. An uneven scale, measuring a.c.
D. An even scale, measuring d.c.
23. One important advantage of an electrostatic meter is that:
- A. It measures very small currents.
B. It will handle large currents.
C. It can detect ac voltages.
D. It draws a large current from the source.
24. What is the percentage error of reading of a voltmeter due to voltmeter loading if it has 50 k Ω internal resistance and a load resistor of 1 k Ω ?
- A. 1.96
B. 2.76
C. 3.5
D. 2.0
25. What type of meter can be used to measure ac currents without breaking open the circuit?
- A. An analog VOM.
B. An amp-clamp probe
C. A DMM.
D. There isn't such a meter.
26. A/an _____ works by directly applying a voltage being measured to an electron beam moving across the oscilloscope screen.
- A. probe
B. analog oscilloscope
C. digital oscilloscope
D. CRT
27. Determine the AC sensitivity of full wave AC voltmeter with a full-scale current of 100 μ A.
- A. 4500
B. 10000
C. 9000
D. 1000
28. What is the method of damping for a moving-iron instrument?
- A. Linear
B. Air
C. Eddy current
D. Hairspring
29. To measure power-supply voltage being used by a circuit, a voltmeter:
- A. Is placed in series with the circuit that works from the supply
B. Is placed between the negative pole of the supply and the circuit working from the supply
C. Is placed between the positive pole of the supply and the circuit working from the supply
D. Is placed in parallel with the circuit that works from the supply.
30. Which of the following measurements is usually the most inconvenient and time-consuming when troubleshooting?
- A. resistance measurements
B. dc voltage measurements
C. current measurements
D. ac voltage measurements.
31. A/an _____, which measures p.d., has a high resistance (ideally infinite) and must be connected in parallel with the part of the circuit whose p.d. is required.
- A. voltmeter
B. ammeter
C. wattmeter
D. ohmmeter
32. Which of the following statements is false?
- A. A digital computer requires a D/A converter
B. When negative feedback is used in an op amp, a constant and predictable voltage gain results



- C. A digital voltmeter requires a D/A converter
D. The value of frequency at which the open-loop gain has fallen to unity is called the transition frequency
33. One advantage of an electromagnet meter over a permanent-magnet meter is that:
- A. The electromagnet meter costs much less.
B. The electromagnet meter need not be aligned with the earth's magnetic field.
C. The permanent-magnet meter has a more sluggish coil.
D. The electromagnet meter is more rugged.
34. Instruments are manufactured that combine a moving coil meter with a number of shunts and series multipliers, to provide a range of readings on a single scale graduated to read current and voltage. If a battery is incorporated then resistance can also be measured. Such instruments are called
- A. Multimeters
B. Universal instruments
C. Multi-range instruments
D. All choices are correct
35. A current meter should have a
- A. very high internal resistance.
B. very low internal resistance.
C. infinitely high internal resistance.
D. None of the above.
36. In the ____ moving-iron instrument, two pieces of iron are placed inside the solenoid, one being fixed, and the other attached to the spindle carrying the pointer. When current passes through the solenoid, the two pieces of iron are magnetized in the same direction and therefore repel each other. The pointer thus moves across the scale.
- A. attraction type
B. repulsion type
C. pivot type
D. moveable type
37. When using an ohmmeter to measure the continuity of a wire, the resistance should measure
- A. about $0\ \Omega$ if the wire is good.
B. infinity if the wire is broken (open).
C. very high resistance if the wire is good.
D. Both A and B.
38. The meter movement with $I_{FS} = 50\ \mu A$ and $R_M = 1\ k\Omega$ is to be converted to an AC voltmeter using half-wave rectification. Compute the size of the series resistor required to have full deflection for an input sine wave of 40 volts rms.
- A. 450 k Ω C. 359 k Ω
B. 254 k Ω D. 300 k Ω
39. Which of the following apply to a moving coil instrument?
- A. An uneven scale, measuring dc.
B. An even scale, measuring a.c.
C. An uneven scale, measuring a.c.
D. An even scale, measuring d.c.
40. When an ammeter is required to measure currents of larger magnitude, a proportion of the current is diverted through a low-value resistance connected in parallel with the meter. Such a diverting resistor is called a ____.
- A. series C. shunt
B. multiplier D. bypass
41. Which of the following would refer to a moving iron instrument?
- A. An uneven scale, measuring dc.
B. An even scale, measuring a.c.
C. An uneven scale, measuring a.c.
D. Both A and C
42. Voltmeter loading is usually a problem when measuring voltages in
- A. parallel circuits.
B. low-resistance circuits.
C. a series circuit with low-resistance values.
D. high-resistance circuits.
43. An ammeter with an internal resistance of $50\ \Omega$ is used to measure a current through a load resistor $R_L = 1\ k\Omega$. Determine the percentage error of the reading due to ammeter insertion.
- A. 67.4 % C. 6.74 %
B. 4.76 % D. 47.6 %
44. ____ are useful whenever two signals are to be compared simultaneously.
- A. Direct voltage measurement oscilloscope
B. Alternating voltage measurement oscilloscope
C. Double beam oscilloscope
D. Waveform point oscilloscope
45. A VU meter is *never* used for measurement of:
- A. Sound C. Power
B. Decibels D. Energy
46. What is the method of control for a moving-coil instrument?
- A. Linear
B. Air
C. Eddy current
D. Hairspring
47. A voltmeter should have a
- A. Resistance of about $0\ \Omega$.
B. very low resistance.
C. very high internal resistance.
D. None of the above.
48. The figure below shows a scale of multi-range ammeter. What is the current indicated when switched to a 25 A scale?



- A. 84 A
- B. 14 A
- C. 5.6 A
- D. 8.4 A

49. Which of the following needed to extend the range of a milliammeter to read voltages of the order of 100 V?

- A. A parallel high-value resistance
- B. A series-value resistance
- C. A parallel low-value resistance
- D. A series low-value resistance

50. A/an _____, which measures current, has a low resistance (ideally zero) and must be connected in series with the circuit.

- A. voltmeter
- B. wattmeter
- C. ammeter
- D. ohmmeter

END